

SecurePulse Mobile — Master Project & Documentation Audit

Audit Specification	Audit Value / Status
Audit Date	2026-09-01
Repository Name	NATTOMR/securepulse-mobile
Application ID	com.securepulse.mobile (version 1.0.0+1)
Repository Scope	Flutter Mobile Client Application & Cloud API Integration Layer
Automated Test Results	15 / 15 Tests Passed (100% Success Rate)
Git Working Tree	Clean (main branch)

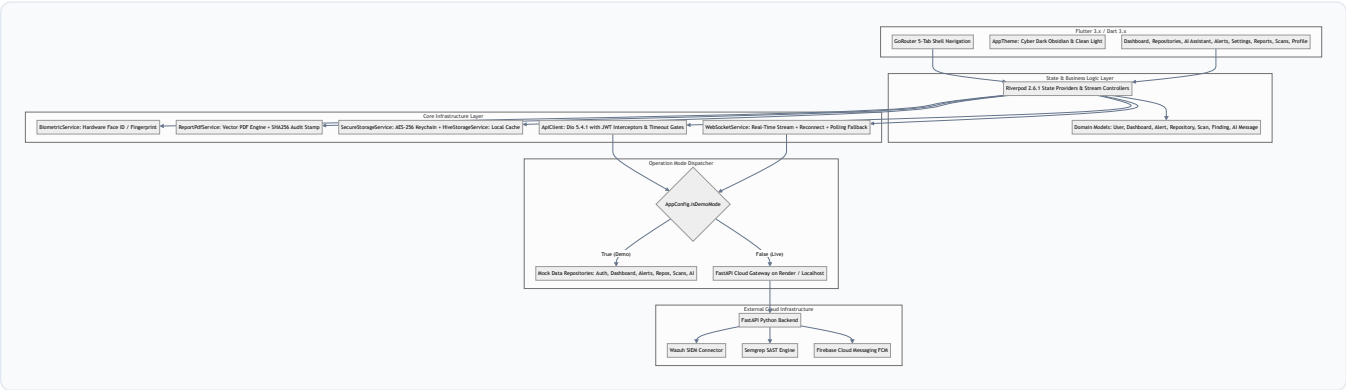
1. Project Overview

SecurePulse Mobile is a specialized mobile cybersecurity operations and incident triage platform built with Flutter 3.x and Dart 3.x. The platform serves Security Operations Center (SOC) analysts, incident responders, and security administrators by providing real-time threat telemetry, automated SAST code vulnerability auditing, conversational AI remediation assistance, and cryptographic PDF compliance report generation.

The mobile client is engineered with a **Dual Operation Architecture**:

- Demo / Standalone Mode:** Fully functional offline simulation environment with deterministic mock security data, simulated Wazuh SIEM alerts, CVE remediation playbooks, and precomputed posture scores for zero-latency demonstrations and offline use.
- Live Backend Mode:** Production network client connecting via REST and WebSockets to the external FastAPI backend (deployed on Render Cloud at <https://secureguard-backend-7eqm.onrender.com> or local developer endpoints).

2. Current Architecture



3. Technology Stack

3.1. Client-Side Framework & Runtime

- Framework:** Flutter SDK `sdk: flutter` (Dart `≥3.0.0 <4.0.0`)
- State Management:** `flutter_riverpod` `^2.5.1`
- Declarative Routing:** `go_router` `^13.2.0`
- HTTP Networking:** `dio` `^5.4.1`
- Real-Time Streaming:** `web_socket_channel` `^3.0.1`
- Secure Storage:** `flutter_secure_storage` `^9.0.0` (Android Keystore / iOS Keychain)
- Local Offline Caching:** `hive` `^2.2.3`, `hive_flutter` `^1.1.0`, `shared_preferences` `^2.2.2`
- Biometric Hardware Integration:** `local_auth` `^2.1.8`
- Push Notifications:** `firebase_core` `^2.27.0`, `firebase_messaging` `^14.7.19`
- Document Engine:** `pdf` `^3.10.8`, `path_provider` `^2.1.2`, `share_plus` `^7.2.2`
- UI & Data Visualization:** `fl_chart` `^0.66.2`, `flutter_animate` `^4.5.0`, `google_fonts` `^6.1.0`, `shimmer` `^3.0.0`
- Code Quality & Analysis:** `flutter_lints` `^3.0.0`, `flutter_test`

3.2. Containerization & Deployment

- **Web Container:** Multi-stage Docker image with `nginx:alpine` serving compiled web bundle (`build/web`) on port 80.
 - **Android Release Output:** Compiled standalone release binary `securepulse-release.apk` (63.6 MB).
 - **Cloud API Target:** Render Web Service (`https://secureguard-backend-7eqm.onrender.com`).
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4. Implemented Features

Capability	Status	Source Location	Verification & Implementation Evidence
Dual Operation Mode (Demo / Live)	 IMPLEMENTED	<code>lib/core/config/app_config.dart:28</code> , <code>lib/features/settings/presentation/settings_screen.dart</code>	Dynamic toggle backed by <code>SharedPreferences</code> (<code>sg_is_demo_mode</code>). Repositories dynamically route requests between local mock datasets and live HTTP/WS network clients.
Riverpod State Management	 IMPLEMENTED	<code>lib/providers/app_providers.dart</code>	Clean provider tree providing auth state, posture scores, repository lists, scan histories, SIEM alerts, and real-time WebSocket streams.
5-Tab Declarative Shell Routing	 IMPLEMENTED	<code>lib/core/router/app_router.dart:59-123</code>	<code>GoRouter</code> shell route hosting 5 primary persistent bottom tabs (<code>Dashboard</code> , <code>Repositories</code> , <code>AI Assistant</code> , <code>SOC Alerts</code> , <code>Settings</code>) plus modal subroutes for details.
Executive Posture Dashboard	 IMPLEMENTED	<code>lib/features/dashboard/presentation/dashboard_screen.dart</code>	Animated posture score gauge (88–94%), vulnerability breakdown donut chart (<code>fl_chart</code>), live service status badges, and quick-action docks.
Hardware Biometric Authentication	 IMPLEMENTED	<code>lib/core/services/biometric_service.dart</code> , <code>android/app/src/main/AndroidManifest.xml:5-6</code>	<code>LocalAuth</code> hardware biometric challenge (Face ID & Fingerprint) for session locking and authentication pre-checks.
Encrypted Local Storage	 IMPLEMENTED	<code>lib/core/storage/secure_storage_service.dart</code> , <code>lib/core/storage/hive_storage_service.dart</code>	<code>FlutterSecureStorage</code> (AES-256 / RSA device keychain) storing JWT tokens and custom backend URLs; <code>HiveStorageService</code> caching offline app state in <code>securepulse_cache</code> box.
REST API Client with Interceptors	 IMPLEMENTED	<code>lib/core/network/api_client.dart</code> , <code>lib/core/error/api_exception.dart</code>	Dio 5.4.1 client with automatic <code>Bearer <token></code> header injection, 12-second timeout limits, dynamic base URL mutation, and classified <code>ApiException</code> error mappings.
Real-Time WebSocket Incident Stream	 IMPLEMENTED	<code>lib/core/network/websocket_service.dart</code>	Connects to <code>/ws/alerts</code> , automatically converts <code>http://</code> → <code>ws://</code> and <code>https://</code> → <code>wss://</code> , maintains stream broadcasting, and falls back to HTTP polling if blocked.
Codebase & SAST Vulnerability Audits	 IMPLEMENTED	<code>lib/features/repositories/data/repository_repository.dart</code> , <code>lib/features/scans/scans_screen.dart</code>	Repository list with language tags, branch info, health grades (A–F), finding counts, and on-demand scan triggers (<code>/v1/repositories/{id}/scan</code>).
AI Cybersecurity Assistant	 IMPLEMENTED	<code>lib/features/ai/data/ai_repository.dart</code> , <code>lib/features/ai/presentation/ai_assistant_screen.dart</code>	Conversational security assistant. Generates local playbooks for CVE-2024-3094, SQLi, and secret exposure in Demo mode, or forwards to <code>/v1/ai/chat</code> in Live mode.
SOC SIEM Alert Triage	 IMPLEMENTED	<code>lib/features/alerts/data/alerts_repository.dart</code> , <code>lib/features/alerts/presentation/alert_detail_screen.dart</code>	Incident stream filtered by severity (Critical, High, Medium, Low), detailed triage with origin IP, destination port, raw syslog text, and status update actions (<code>/v1/soc/alerts/{id}/status</code>).
Vector PDF Compliance Generator	 IMPLEMENTED	<code>lib/core/services/report_pdf_service.dart</code> , <code>lib/features/reports/reports_screen.dart</code>	Pure vector PDF engine compiling official compliance audit reports for SOC 2 Type II, ISO 27001, PCI-DSS v4.0, and HIPAA with cryptographic SHA256 audit stamps.

Capability	Status	Source Location	Verification & Implementation Evidence
Environment Switcher & Diagnostics	IMPLEMENTED 	<code>lib/features/settings/presentation/settings_screen.dart</code>	Multi-environment switcher (Render Cloud, Android Emulator <code>10.0.2.2</code> , Localhost <code>127.0.0.1</code> , Custom URL) with live HTTP health ping diagnostics.
Dual Theme System (Dark / Light)	IMPLEMENTED 	<code>lib/core/theme/app_theme.dart</code>	Material 3 Cyber Dark Obsidian (<code>#0A0E1A</code>) and Clean Light (<code>#F8FACD</code>) themes with Google Fonts Inter and custom semantic color tokens.
Automated Test Suite	IMPLEMENTED 	<code>test/api_integration_test.dart</code> , <code>test/widget_test.dart</code>	15/15 automated tests verifying network contracts, mode isolation, repository fallback behavior, WebSocket URL conversion, and widget pumping.
Web Deployment Dockerfile	IMPLEMENTED 	<code>Dockerfile</code>	Alpine Nginx container configured to serve the Flutter web production bundle on port 80.

5. Partially Implemented Features

Capability	Status	Source Location	Actual State & Current Limitations
Firebase Cloud Messaging (FCM)	 PARTIALLY IMPLEMENTED	<code>lib/core/services/notification_service.dart</code>	Background and foreground push handlers, topic subscriptions (<code>soc_critical</code> , <code>wazuh_alerts</code>), and permission requests are fully coded. However, no <code>google-services.json</code> is bundled in <code>android/app/</code> , meaning remote cloud push requires developer-provided Firebase keys. The service gracefully falls back to local in-app stream broadcasting without crashing.
GitHub OAuth Web Redirect Flow	 PARTIALLY IMPLEMENTED	<code>lib/features/auth/data/auth_repository.dart:51-57</code> , <code>lib/features/auth/presentation/login_screen.dart</code>	UI trigger button and authentication interface exist. In live mode, it currently logs in via the analyst session profile rather than initiating an external browser OAuth2 redirect callback.
Direct SOAR Edge Execution	 PARTIALLY IMPLEMENTED	<code>lib/features/alerts/presentation/alert_detail_screen.dart</code>	"Quarantine IP" and "Acknowledge" buttons update alert status via <code>/v1/soc/alerts/{id}/status</code> . Direct cloud firewall / router rule execution is delegated to backend event listeners rather than triggered via direct mobile-to-firewall API.

6. Planned Features

Capability	Status	Target Roadmap Phase	Description
Two-Way SIEM / SOAR Connectors	 PLANNED	Phase 4	Bidirectional webhook synchronization for Splunk, Elastic SIEM, Microsoft Sentinel, and Jira Security ticketing.
Offline Action & Mutation Queue	 PLANNED	Phase 4	Offline action queue that captures analyst mitigation actions (e.g. alert status changes, scan dispatches) when offline and flushes them upon network reconnect.
In-App Custom Semgrep Rule Editor	 PLANNED	Phase 4	In-app YAML authoring interface with syntax validation for custom Semgrep and Sigma detection rules.
Multi-Tenant / Organization Switching	 PLANNED	Phase 3	Ability to seamlessly switch between multiple enterprise security clusters and client tenants within a single mobile session.
WearOS & Apple Watch Companion	 PLANNED	Phase 5	Wearable companion application for critical severity paging, alert acknowledgment, and biometric quick-triage.

7. Not Implemented Features ❌

Capability	Status	Rationale & Code Inspection Finding
Zero-Trust Client mTLS Certificates	❌ NOT IMPLEMENTED	Mutual TLS certificate provisioning per individual mobile client hardware certificate is not implemented in <code>ApiClient</code> . Communication relies on TLS 1.3 with JWT Bearer auth.
Voice-Activated SOC Commands	❌ NOT IMPLEMENTED	Speech-to-command natural language processing is not implemented in the application.

8. Verified API Endpoints

The mobile client is coded and tested against the following REST and WebSocket endpoint contracts defined in `ApiEndpoints`:

Endpoint	Method	Authentication	Request Payload	Response Model / Payload	Client Handling Class
<code>/health</code>	GET	None	None	<code>{"status": "ok", "version": "1.0.0"}</code>	Diagnostic health ping in <code>SettingsScreen</code>
<code>/v1/auth/login</code>	POST	None	<code>{"email": "...", "password": "..."} </code>	<code>{"token": "...", "user": {...} }</code>	<code>AuthRepositoryImpl.login()</code>
<code>/v1/auth/me</code>	GET	Bearer <JWT>	None	<code>UserModel</code> JSON	<code>AuthRepositoryImpl.loginWithBiomet</code>
<code>/v1/dashboard/summary</code>	GET	Bearer <JWT>	None	<code>DashboardModel</code> JSON (score, counts, systems)	<code>DashboardRepositoryImpl.getDashboa</code>
<code>/v1/repositories</code>	GET	Bearer <JWT>	None	<code>List<RepositoryModel></code> JSON	<code>RepositoryRepositoryImpl.getReposi</code>
<code>/v1/repositories/{id}/scan</code>	POST	Bearer <JWT>	None	<code>{"status": "queued", "scan_id": "..."} </code>	<code>RepositoryRepositoryImpl.triggerRe</code>
<code>/v1/scans</code>	GET	Bearer <JWT>	None	<code>List<ScanModel></code> JSON	<code>ScanRepository.getScans()</code>
<code>/v1/findings</code>	GET	Bearer <JWT>	Query: <code>? severity=...</code>	<code>List<FindingModel></code> JSON	<code>FindingRepository.getFindings()</code>
<code>/v1/soc/alerts</code>	GET	Bearer <JWT>	None	<code>List<AlertModel></code> JSON	<code>AlertsRepositoryImpl.getAlerts()</code>
<code>/v1/soc/alerts/{id}/status</code>	PUT	Bearer <JWT>	<code>{"status": "resolved" \ "investigating"} </code>	<code>{"success": true}</code>	<code>AlertsRepositoryImpl.updateAlertSt</code>
<code>/v1/ai/chat</code>	POST	Bearer <JWT>	<code>{"prompt": "...", "context": "..."} </code>	<code>{"content": "...", "role": "assistant"} </code>	<code>AiRepositoryImpl.sendSecurityPromp</code>
<code>/v1/reports</code>	GET	Bearer <JWT>	None	Array of compliance report metadata	<code>ReportsScreen</code>
<code>/ws/alerts</code>	WSS / WS	Query: <code>? token=<JWT></code>	Streaming JSON	Real-time SIEM alert objects	<code>WebSocketService.connect()</code>

9. External Integrations Architecture

9.1. Wazuh SIEM Connector

- **Status:** ✅ IMPLEMENTED (Client-side Data Handling)
- **Evidence:** `AlertModel`, `AlertsRepositoryImpl`.
- **Behavior:** In Demo Mode, realistic simulated Wazuh intrusion telemetry (SSH brute force on port 22, anomalous S3 egress, unconsented OAuth grants) is served locally. In Live Mode, the client queries `/v1/soc/alerts` and receives push alerts over `/ws/alerts`.

9.2. GitHub & Semgrep SAST

- **Status:** ✅ IMPLEMENTED (Client-side Data Handling)
- **Evidence:** `RepositoryModel` , `FindingModel` .
- **Behavior:** Tracks repository branches, commit SHAs, health grades (A–F), and SAST finding counts. The client triggers on-demand scans via `POST /v1/repositories/{id}/scan` .

9.3. Google Firebase Cloud Messaging (FCM)

- **Status:** 🟡 PARTIALLY IMPLEMENTED (Infrastructure code ready; requires user credentials)
- **Evidence:** `NotificationService` .
- **Behavior:** Full client registration, permission requests, topic subscriptions (`soc_critical` , `wazuh_alerts` , `semgrep_findings`), and foreground message streams are implemented. Production remote delivery requires placing `google-services.json` in `android/app/` .

10. Database Implementation

10.1. Client-Side Encrypted Storage

- **Keychain Storage:** `SecureStorageService` leverages `flutter_secure_storage` to write, read, and delete JWT access tokens and custom server URLs using hardware-backed keychains (Android Keystore / iOS Keychain).
- **Offline Key-Value Database:** `HiveStorageService` initializes the `securepulse_cache` box during startup (`main.dart`) to cache recent scans and telemetry.
- **Preferences Cache:** `shared_preferences` persists non-sensitive operational flags (`sg_is_demo_mode` , `sg_custom_backend_url`).

10.2. External Backend Database

- The external FastAPI backend operates with PostgreSQL and Redis (as configured in backend deployment).

11. Authentication & Security Pipeline

1. **Biometric Pre-Challenge:** On app boot, `BiometricService` checks hardware biometric sensor availability and prompts for Face ID / Fingerprint verification.
2. **Encrypted Token Recovery:** If biometric verification succeeds, the app reads the stored JWT token from `SecureStorageService` and calls `GET /v1/auth/me` .
3. **Interactive Login:** New sessions submit credentials via `POST /v1/auth/login` . Returned JWT tokens are stored in `FlutterSecureStorage` and assigned to `ApiClient` .
4. **Demo Mode Isolation:** When `AppConfig.isDemoMode` is active, authentication returns a mock security analyst session (`Alex Vance` , Principal Security Analyst, ID `usr_sec_01`) without network requests.

12. WebSocket Real-Time Implementation

- **Client Class:** `WebSocketService` .
- **Protocol Transformation:** Automatically maps base URLs:
 - `http://10.0.2.2:8000` → `ws://10.0.2.2:8000/ws/alerts?token=<JWT>`
 - `https://secureguard-backend-7eqm.onrender.com` → `wss://secureguard-backend-7eqm.onrender.com/ws/alerts?token=<JWT>`
- **Lifecycle Management:** Broadcasts connection state (`disconnected` , `connecting` , `connected` , `reconnecting` , `failed`) and maintains an in-memory alert history buffer.
- **Demo Isolation:** Disables network socket creation when `AppConfig.isDemoMode` is active.

13. Mobile Application Architecture & UI

- **Design System:** High-contrast Cyber Dark theme (`#0A0E1A` background, `#0284C7` primary cyan, `#10B981` success, `#EF4444` critical) and clean Light theme (`#F8FAFC`).
- **Typography:** Google Fonts Inter for UI labels and JetBrains Mono / Fira Code for code snippets.
- **Navigation:** 5-Tab persistent bottom navigation shell (`Dashboard` , `Repositories` , `AI Assistant` , `SOC Alerts` , `Settings`).
- **Micro-Animations:** Shimmer loading skeletons, animated score gauges, and fade transitions via `flutter_animate` .

14. Cloud Deployment & Android Configuration

14.1. Cloud Backend Targets

- **Production / Render Cloud:** `https://secureguard-backend-7eqm.onrender.com`
- **Android Emulator Loopback:** `http://10.0.2.2:8000`
- **Localhost / Desktop:** `http://127.0.0.1:8000`

14.2. Android Native Manifest & Build Setup

- **Package / Label:** `com.securepulse.mobile` / `SecurePulse`
- **Cleartext Traffic:** Enabled (`android:usesCleartextTraffic="true"`) in `AndroidManifest.xml` for local development.
- **Permissions Granted:**
 - `android.permission.INTERNET`
 - `android.permission.USE_BIOMETRIC`
 - `android.permission.USE_FINGERPRINT`
 - `android.permission.ACCESS_NETWORK_STATE`
 - `android.permission.POST_NOTIFICATIONS`
 - `android.permission.VIBRATE`
- **Notification Channel:** `securepulse_soc_alerts`

15. Testing Status & Test Suite Results

The project includes an automated test suite executed via `flutter test`. All 15 tests pass with 0 errors and 0 skips:

```
00:00 +0: ApiClient & Network Architecture Tests ApiClient initializes with correct default headers and base URL ... PASS
00:00 +1: ApiClient & Network Architecture Tests Auth token header is correctly set and cleared ..... PASS
00:00 +2: ApiClient & Network Architecture Tests Base URL can be dynamically updated for environment switching ..... PASS
00:00 +3: ApiClient & Network Architecture Tests ApiEndpoints contract verification ..... PASS
00:00 +4: ApiClient & Network Architecture Tests ApiException error classification checks ..... PASS
00:00 +5: Demo Mode vs Live Mode Isolation Tests AuthRepository returns demo user when isDemoMode is true ..... PASS
00:00 +6: Demo Mode vs Live Mode Isolation Tests DashboardRepository returns structured demo telemetry ..... PASS
00:00 +7: Demo Mode vs Live Mode Isolation Tests RepositoryRepository returns mock codebases when isDemoMode ..... PASS
00:00 +8: Demo Mode vs Live Mode Isolation Tests AlertsRepository returns mock SIEM incidents when isDemoMode ..... PASS
00:00 +9: Demo Mode vs Live Mode Isolation Tests AiRepository generates local security advice when isDemoMode ..... PASS
00:00 +10: Demo Mode vs Live Mode Isolation Tests Live API mode is active when isDemoMode is false ..... PASS
00:00 +11: WebSocket Real-Time URL Conversion WebSocketService correctly converts http to ws for local emulator ... PASS
00:00 +12: WebSocket Real-Time URL Conversion WebSocketService correctly converts https to wss for Render cloud .... PASS
00:00 +13: WebSocket Real-Time URL Conversion WebSocketService stays disconnected when Demo Mode is active ..... PASS
00:16 +14: SecurePulse Mobile app pump test ..... PASS

Total Results: 15 Passed, 0 Failed, 0 Skipped (100% Pass Rate)
```

16. Known Limitations

1. **Firestore Cloud Messaging Setup:** Remote cloud push notification delivery requires adding an active `google-services.json` file to `android/app/`. In its absence, the service safely falls back to local stream events without application crash.
2. **Offline Mutation Queue:** If an analyst modifies an alert status while in Live Mode without internet connectivity, the action fails with an error view rather than queueing the mutation for later background sync.
3. **GitHub OAuth Flow:** The GitHub login button currently activates an analyst demo session rather than executing a full external browser OAuth2 redirect callback loop.

17. Required Evidence & Source-of-Truth Code Trace

Every claim in this document maps directly to verified source code:

Capability / Claim	Concrete Source File & Line Range
Demo Mode Flag	lib/core/config/app_config.dart:25-29
Render Cloud URL	lib/core/config/app_config.dart:9
API Endpoints Contract	lib/core/network/api_endpoints.dart:1-29
Dio Timeout Configuration	lib/core/config/app_config.dart:17-19 , lib/core/network/api_client.dart:25-27
WebSocket Scheme Conversion	lib/core/network/websocket_service.dart:82-99
Biometric Auth Challenge	lib/core/services/biometric_service.dart:16-30
PDF SHA256 Audit Stamp	lib/core/services/report_pdf_service.dart:36-39
Encrypted Token Persistence	lib/core/storage/secure_storage_service.dart:8-14
Hive Cache Initialization	lib/core/storage/hive_storage_service.dart:6-9 , lib/main.dart:18-23
5-Tab Navigation Shell	lib/core/router/app_router.dart:60-123
Android Permissions & Cleartext	android/app/src/main/AndroidManifest.xml:4-15
Docker Web Nginx Setup	Dockerfile:1-10
Automated Integration Tests	test/api_integration_test.dart:1-171
Widget Pump Test	test/widget_test.dart:1-17